Haddock (*Melanogrammus aeglefinus*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Range Score	Range LH DQ	Range Tally	Gulf of Maine Score	Gulf of Maine LH DQ	Gulf of Maine Tally	Georges Bank Score	Georges Bank LH DQ	Georges Bank Tally
Adult Mobility	2.08	2.8		2	2.8		2.16	2.8	
Complexity in Reproductive Strategy	2.96	2.8		2.96	2.8		2.96	2.8	
Dispersal of Early Life Stages	1.74	2.9		1.64	2.8		1.84	3	
Egg and Larval Survival and Settlement Requirements	2.98	2.7		2.96	2.7		3	2.7	
Historical Variability in Temperature	1.88	2.9		1.88	2.9		1.88	2.9	
Juvenile Habitat Requirements	1.3	3		1.4	3		1.2	3	
Other Stressors	1.24	2.7		1.24	2.7		1.24	2.7	
Population Growth Rate	2.16	2.6		2.16	2.6		2.16	2.6	
Prey Specificity	1.32	2.8		1.32	2.8		1.32	2.8	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.36	2.7		1.36	2.7		1.36	2.7	
Spawning Cycle	2.48	2.95		2.52	2.9		2.44	3	
Stock Size/Status	1.46	3		1.2	3		1.72	3	
Total Sensitivity | Certainty	**Moderate (2) | 0.79**			**Moderate (2) | 0.71**			**Moderate (2) | 0.68**		

Exposure

Exposure Factors	Range Mean	Range SD	Range Tally	Gulf of Maine Mean	Gulf of Maine SD	Gulf of Maine Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally
Bottom Temperature	2.34	1.08		2.14	1.05		2.74	1.03	
Bottom Oxygen	2.82	0.97		2.9	0.87		2.9	1.04	
Bottom Salinity	2.54	1.07		2.21	0.98		2.84	1.08	
Bottom Aragonite Solubility	1.24	0.63		1.17	0.43		1.23	0.68	
Net primary production	1.03	0.08		1.06	0.10		1	0.01	
Downward Flux of Particulate Organic Carbon	1.37	0.33		1.47	0.37		1.23	0.22	
Total Exposure	**Moderate (1.94)**	**0.94**		**Moderate (1.75)**	**0.88**		**Moderate (2.24)**	**0.96**	

Vulnerability

Vulnerability	Range Mean	Range SD	Range Tally	Gulf of Maine Mean	Gulf of Maine SD	Gulf of Maine Tally	Georges Bank Mean	Georges Bank SD	Georges Bank Tally
Total Vulnerability	**Moderate (3.88)**	**1.89**		**Low (3.49)**	**1.76**		**Moderate (4.48)**	**1.91**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Haddock. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 6 and Figure 5). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Moderate	Increase
Exposure	High	Moderate	Decrease
Vulnerability	Low	Moderate	Increase
Directional Effect	Negative	Negative-Neutral	None-Increase

Stocks

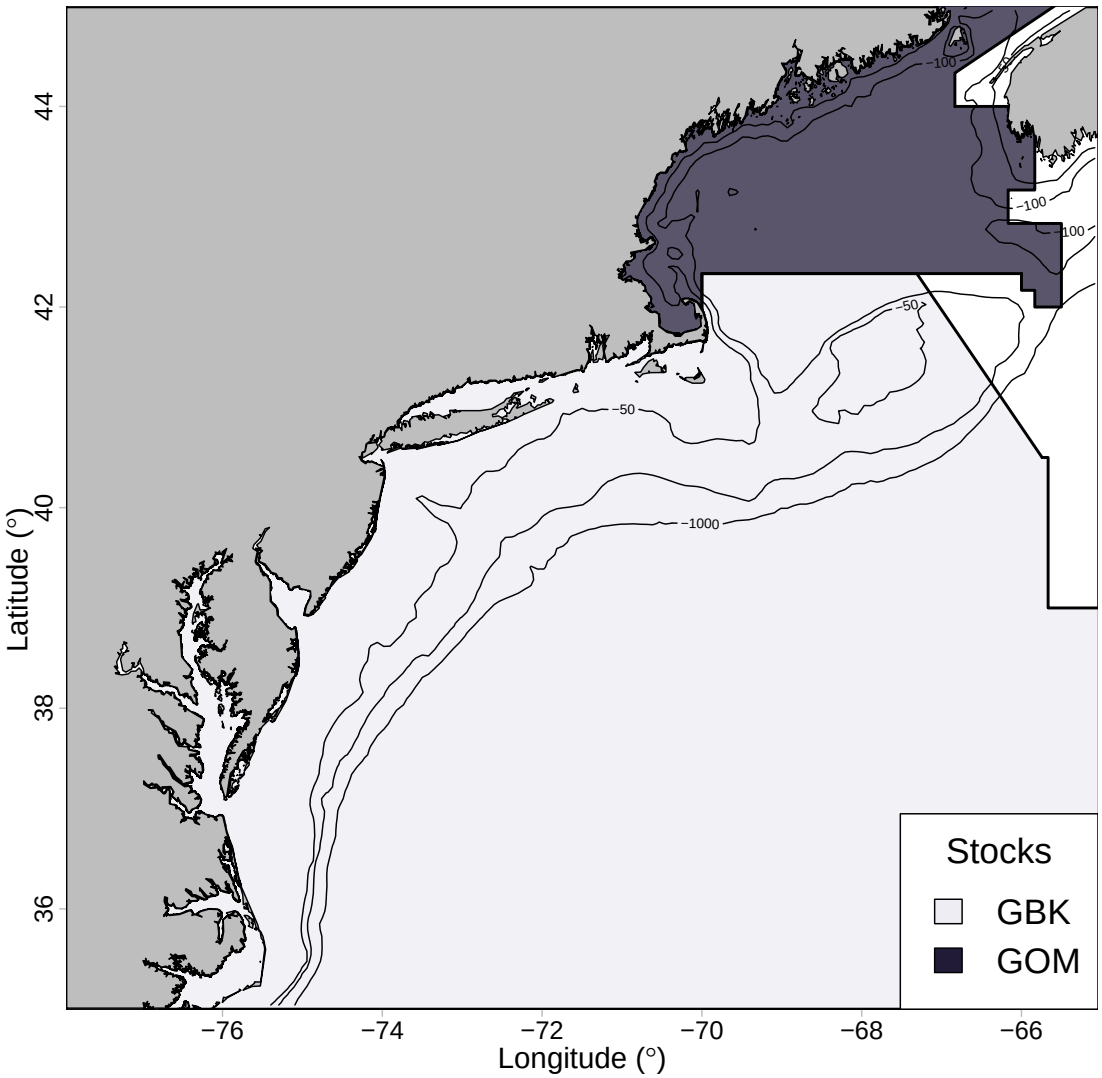


Figure 2: Haddock Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

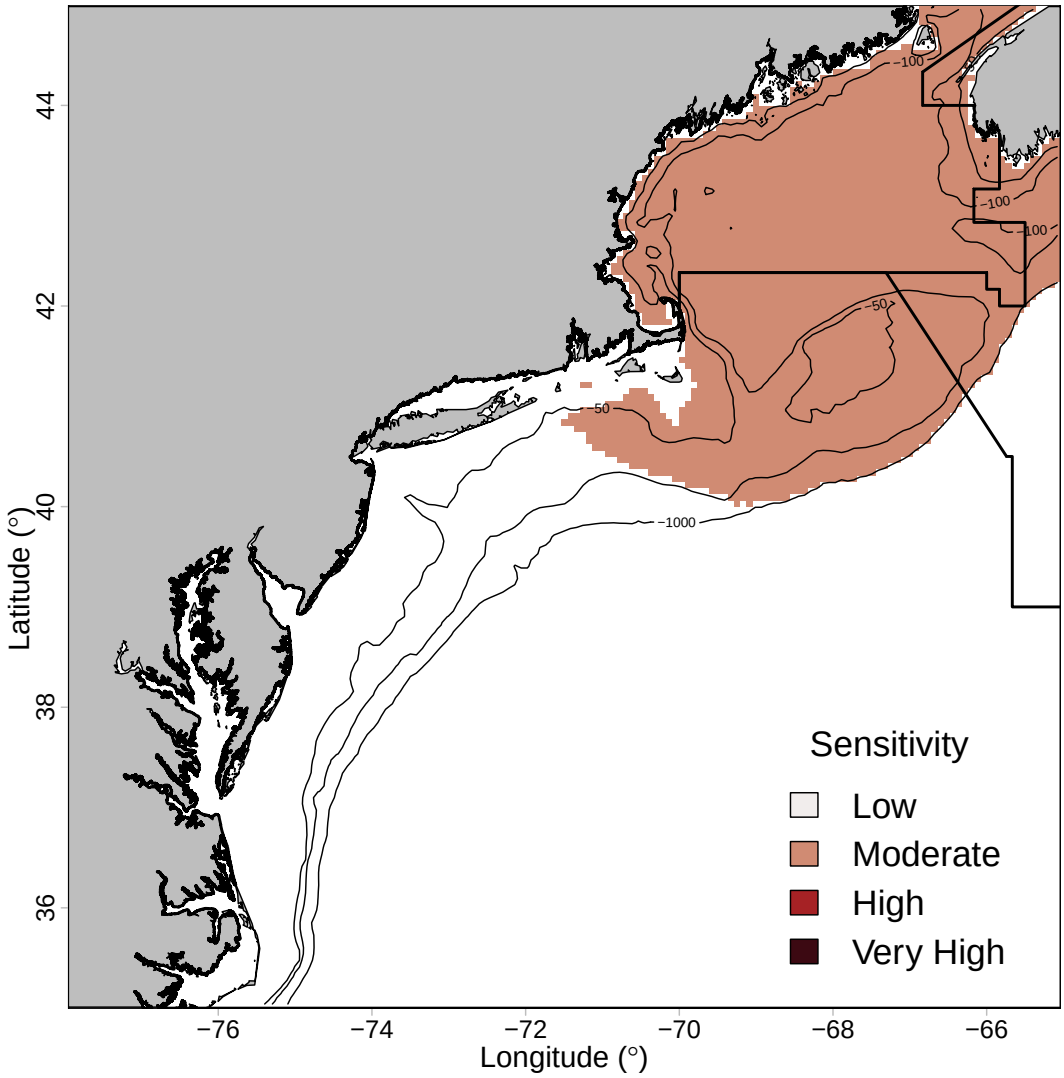


Figure 3: Haddock Sensitivity on the Northeast Shelf

Both the Georges Bank and Gulf of Maine stocks of Haddock received **Moderate** sensitivity scores. Certainties were 68% and 71%, respectively. For both stocks, Complexity in Reproductive Strategy (3) and Egg and Larval Survival and Settlement Requirements (3) had scores greater than 2.5. For the Gulf of Maine stock, Spawning Cycle (2.5) also had a score greater than 2.5 (Figure 1).

Haddock spawn in aggregations. While bottom type conditions drive where haddock spawn, the timing of spawning has been correlated to temperatures in some studies. Retention of larvae is important, especially to the Georges Bank stock, as strong year-classes are present when circulation patterns promote retention on or near Georges Bank (Brodziak 2005).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Haddock received a Low sensitivity score. No attributes scored above 2.5 in the previous assessment (Hare et al. 2016).

Exposure:

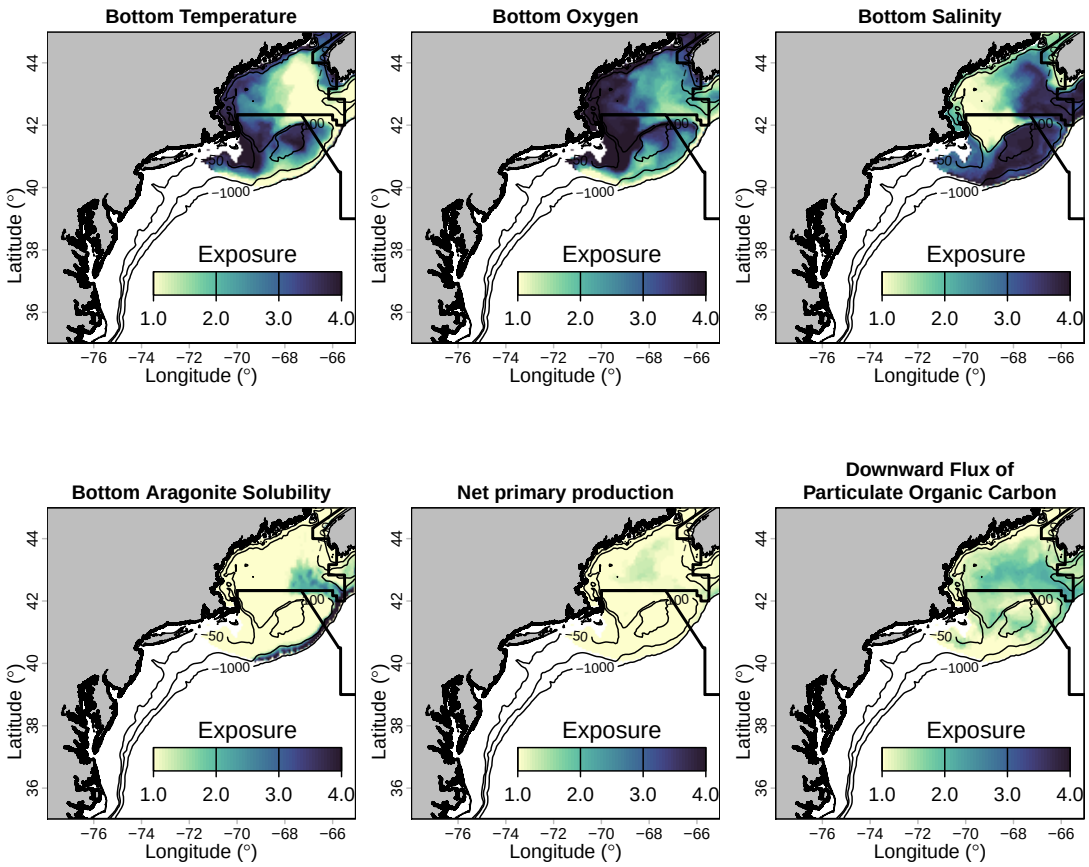


Figure 4: Maps of each Exposure Factor for Haddock. Exposure was calculated between 2014-2023 and 2025-2035

The overall mean Exposure for Haddock across its range was **Moderate** (1.94 +/- 0.94) (Figure 1). Scores were also Moderate in both the Gulf of Maine (1.75 +/- 0.88) and Georges Bank (2.24 +/- 0.96) stocks (Figure 1). Exposure was High along the coast of the Gulf of Maine, south of Cape Cod, and over Georges Bank, with areas of Very High Exposure over the Bank (Figure 5). These patterns were driven by high exposure to bottom temperature, oxygen, and salinity changes (Figure 4).

In the previous Assessment, Haddock received a High Exposure score, driven by sea surface temperatures and ocean acidification (Hare et al. 2016).

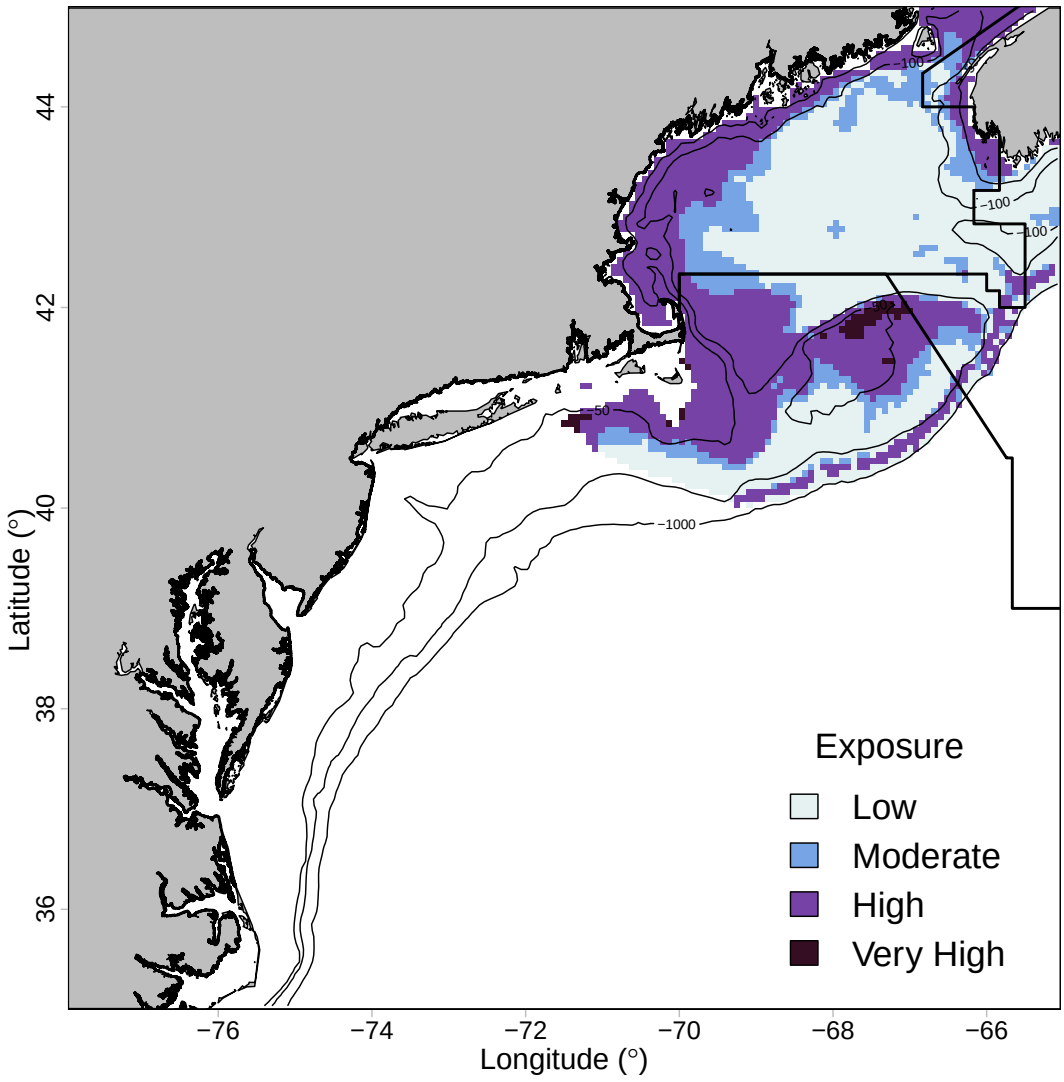


Figure 5: Total Exposure on the Northeast Shelf for Haddock. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 4)

Vulnerability:

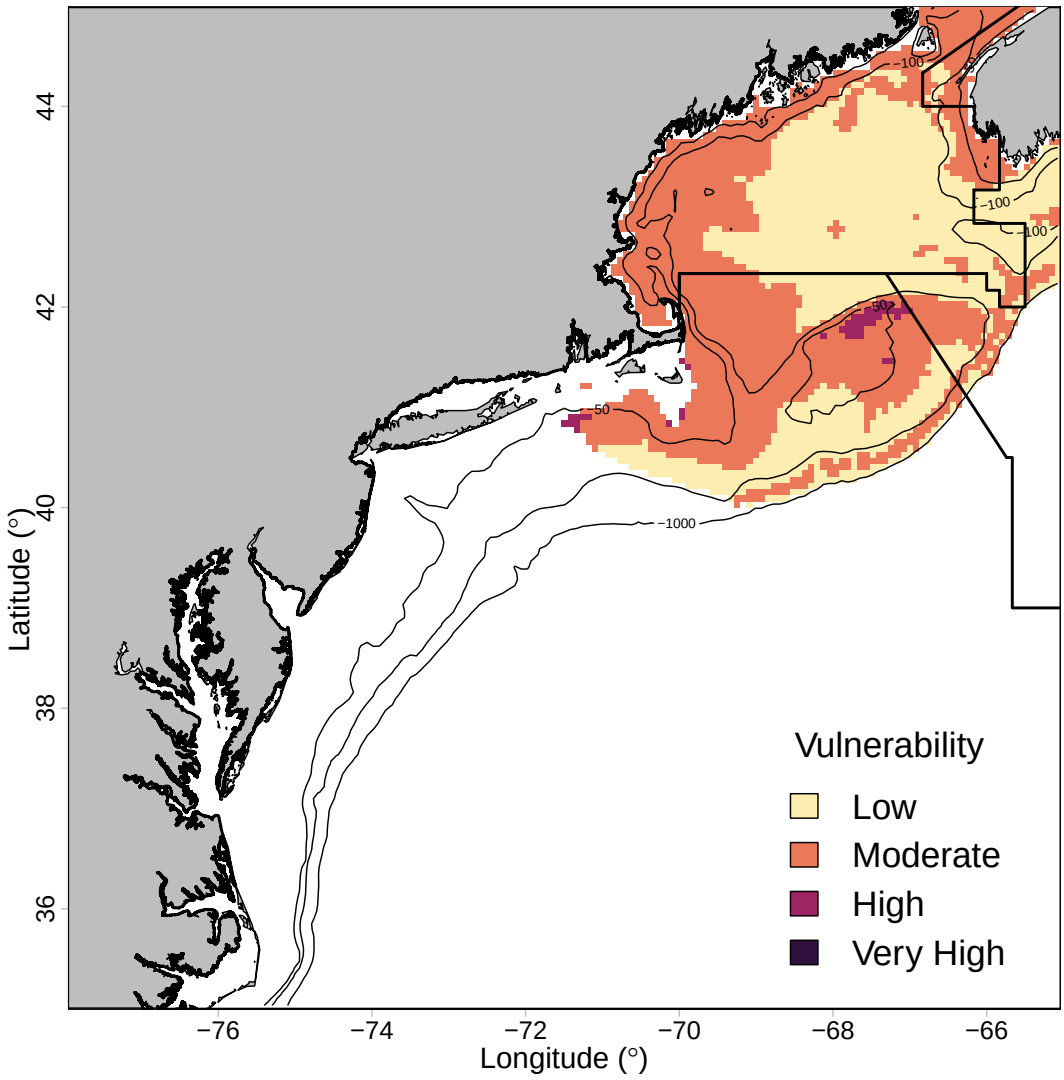


Figure 6: Vulnerability on the Northeast Shelf for Haddock. Vulnerability is the product of sensitivity and exposure (see Figure 5 and Figure 3)

Mean Vulnerability for Haddock was **Moderate** across its range (3.88 +/- 1.89) (Figure 1). Vulnerability was **Low** within the **Gulf of Maine** stock and **Moderate** within the **Georges Bank** stock (3.49 +/- 1.76 and 4.48 +/- 1.91, respectively) (Figure 1). Vulnerability was High over Georges Bank, and Moderate along the coast of Southern New England and the Gulf of Maine (Figure 6).

In the previous Assessment, Haddock received a Low Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

97% of data quality scores were greater than or equal to 2 for both stocks, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The haddock ensemble model received a model confidence score of 2.3 (+/- 0.2), indicating moderate confidence in the model. The modeled distributions aligned with experts' expectations. There were some concerns about the model residuals which may be tied to changes in abundance over time. The ensemble model AUC for Haddock was 0.91, indicating that the model had excellent predictive power.

Additional Metrics

Distribution Shifts:

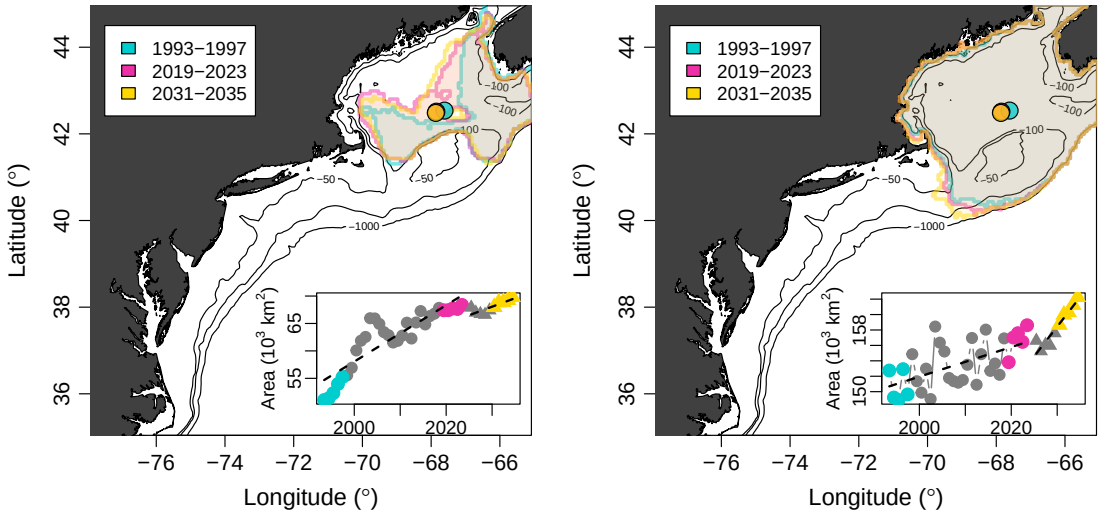


Figure 7: Distribution shifts on the Northeast Shelf for Haddock. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Haddock core habitat area and range significantly increased between 1993 and 2023. Core habitat expanded westward into the Gulf of Maine and the species' total range expanded south and west towards southern New England (Figure 7). The weighted centroid shifted 20 km west during this time.

Under near future climate conditions, Haddock habitat is predicted to continue to grow. Habitat is predicted to continue to expand further into the Gulf of Maine and into southern New England waters (Figure 7). The distribution centroid is predicted to shift a further 3 km south-southwest.

In the previous assessment, haddock received a High Distributional Vulnerability Rank due to their mobility, habitat preferences, and high larval dispersion (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	53,027	150,720	3.43	4.73
2019-2023	19.7 km W	67,784 (+14,756)	156,737 (+6,017)	3.55 (+0.12)	4.73 (0)
2031-2035	3.4 km SSW	68,955 (+1,171)	160,491 (+3,754)	3.62 (+0.06)	4.81 (+0.08)

Directional Effect:

Experts predicted that the **Georges Bank** Haddock stock will be **negatively** impacted by climate change, while climate change will **neutrally** impact the **Gulf of Maine** stock. Certainties were 70% and 68%, respectively. Haddock abundance on Georges Bank has increased recently and coincides with rising ocean temperatures (Friedland et al. 2008) (Wang, Gao, and McCurdy 2024), but increasing temperatures may also lead to increased predation by cod (Durant et al. 2020). Retention plays a large role in recruitment on Georges Bank, but not much is known about the role retention plays in the Gulf of Maine stock (Brodziak 2005).

Experts predicted that haddock would be negatively impacted by climate change in the previous Assessment (Hare et al. 2016).

Climate Effects on Abundance and Distribution

In the Northeast Atlantic, early life stage haddock and cod compete for food and later life stage cod prey upon haddock. Changing climate and the increasing number of warm years increases cod abundance, causing a decline in haddock (Durant et al. 2020). Haddock abundance on Georges Bank has increased significantly in recent years and may coincide with a rise in extreme ocean temperatures or phytoplankton blooms (Friedland et al. 2008) (Wang, Gao, and McCurdy 2024). A large recruitment event in Georges Bank haddock in 2003 was hypothesized to be related to the timing of the fall bloom and enhanced adult condition, affecting reproductive output (Friedland et al. 2008). Spatial distribution changes of haddock were modeled using bottom trawl survey data, and results showed range expansion and contraction in both spring and fall. Higher temperatures correspond with northward distribution shifts on Georges Bank (Wang, Gao, and McCurdy 2024). Future warming may shift haddock's southern range boundary, making it difficult to define stock boundaries.

Life History Synopsis

In the Northwest Atlantic, haddock are found from Newfoundland to Cape May, New Jersey but are most abundant in the Gulf of Maine and on Georges Bank (Brodziak 2005). The stock is managed in two subunits for the Gulf of Maine and Georges Bank (Northeast Fisheries Science Center 2024a) (Northeast Fisheries Science Center 2024b). Haddock primarily spawn on the Northeast peak of Georges Bank and in the Great South Channel. Spawning occurs from January to July with a delay in peak spawning time with increasing latitude (Smith and Morse 1985). Spawning distribution is influenced more by restrictive spawning area and bottom type conditions than by temperature variation (Page and Frank 1989). At water temperatures typical of Georges Bank, haddock eggs hatch in about 15 days. Larvae metamorphose into juveniles in roughly 30-42 days at lengths of 2-3 cm (Page and Frank 1989). Small juveniles initially live and feed in the epipelagic zone and remain in the upper part of the water column for 3-5 months (Fahay 1983). The retention of haddock larvae in suitable nursery areas is an important factor in determining year class strength of Georges Bank haddock. The clockwise gyre around the main portion of Georges Bank provides a physical mechanism to retain haddock larvae on the Bank. There is limited information on retention of larval haddock in the Gulf of Maine. Larvae can potentially be advected as far south as the Mid-Atlantic Bight, but local retention or transport to nursery habitats to the southwest of Georges Bank are important to strong recruitment (Brodziak 2005). Haddock larvae are passive foragers on less motile prey: invertebrate eggs, copepods, and phytoplankton (Marak 1960). Planktonic prey declines after juveniles become demersal, feeding on ophiuroids and polychaetes instead of copepods and pteropods (Brodziak 2005). Common predators of juvenile haddock include spiny dogfish, skates, and other groundfish. Gray seals also prey on adult haddock (Brodziak 2005). Adult haddock are indiscriminate consumers of benthic, sedentary or slow-moving invertebrates: crustaceans, annelids, polychaetes, mollusks, and echinoderms. Prey consumption varies by area and prey availability indicating adaptability (Tam et al. 2016). Juvenile and adult haddock are demersal and usually found between 40-150m depth. Preferred bottom types include gravel, pebbles, clay and smooth hard sand, particularly smooth areas between rocky patches (Klein-MacPhee 2002). Age at maturity ranges from 1-4 years with a maximum age of 22 years (Northeast Fisheries Science Center 2014). The Georges Bank and Gulf of Maine haddock stocks are managed by the New England Fishery Management Council under the Northeast Multispecies (Groundfish) Fishery Management Plan. Both stocks are not overfished and overfishing is not occurring (Northeast Fisheries Science Center 2024a) (Northeast Fisheries Science Center 2024b).

Supplemental Scores

Exposure Timeseries

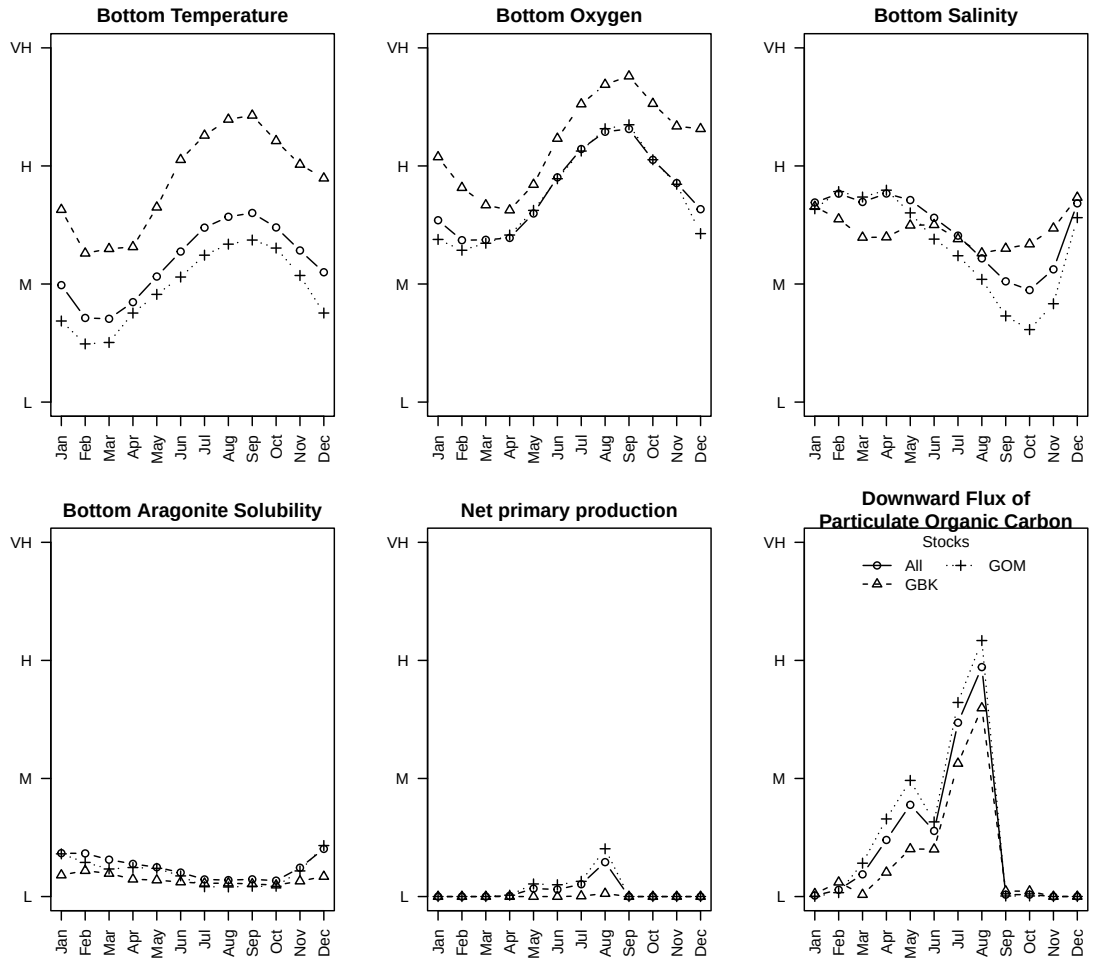


Figure 8: Monthly Timeseries of each Exposure Factor for Haddock. Exposure was calculated between 2014-2023 and 2025-2035

Exposure ranged from Low to Moderate across the range of Haddock, oscillating with changes in bottom temperature, bottom oxygen, and downward carbon flux exposure (Figure 9, Figure 8). Within both stocks, exposure ranged from Low to High. Exposure was High from June through November over Georges Bank, while it was only High in August in the Gulf of Maine stock (Figure 9). This was because bottom temperature and oxygen exposure were greater over Georges Bank than they were in the Gulf of Maine (Figure 8).

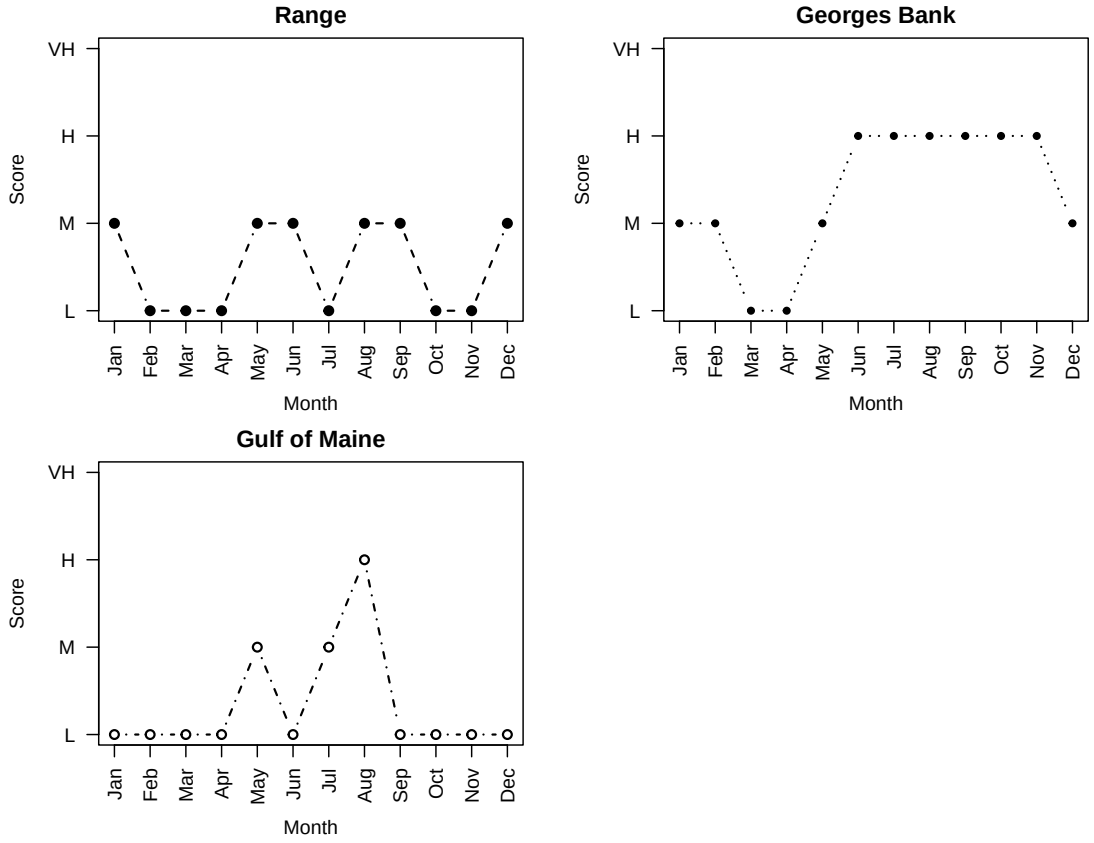


Figure 9: Timeseries of Total Exposure for Haddock

Exposure due to Significant Factors

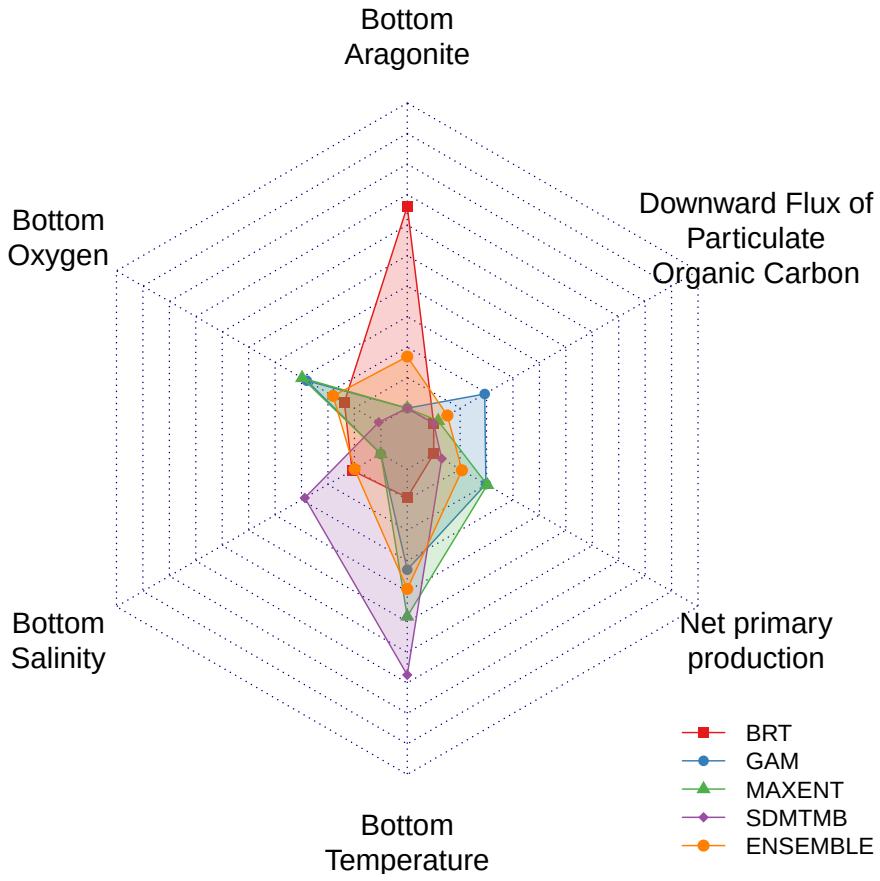


Figure 10: Dynamic Factors driving Haddock distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, aragonite solubility, and net primary productivity were important dynamic drivers of habitat distributions (Figure 10). Haddock are demersal fish and are generalist feeders (Tam et al. 2016). When only these factors were used to calculate Exposure, scores remained Moderate and spatial patterns were similar (Figure 11; Figure 12).

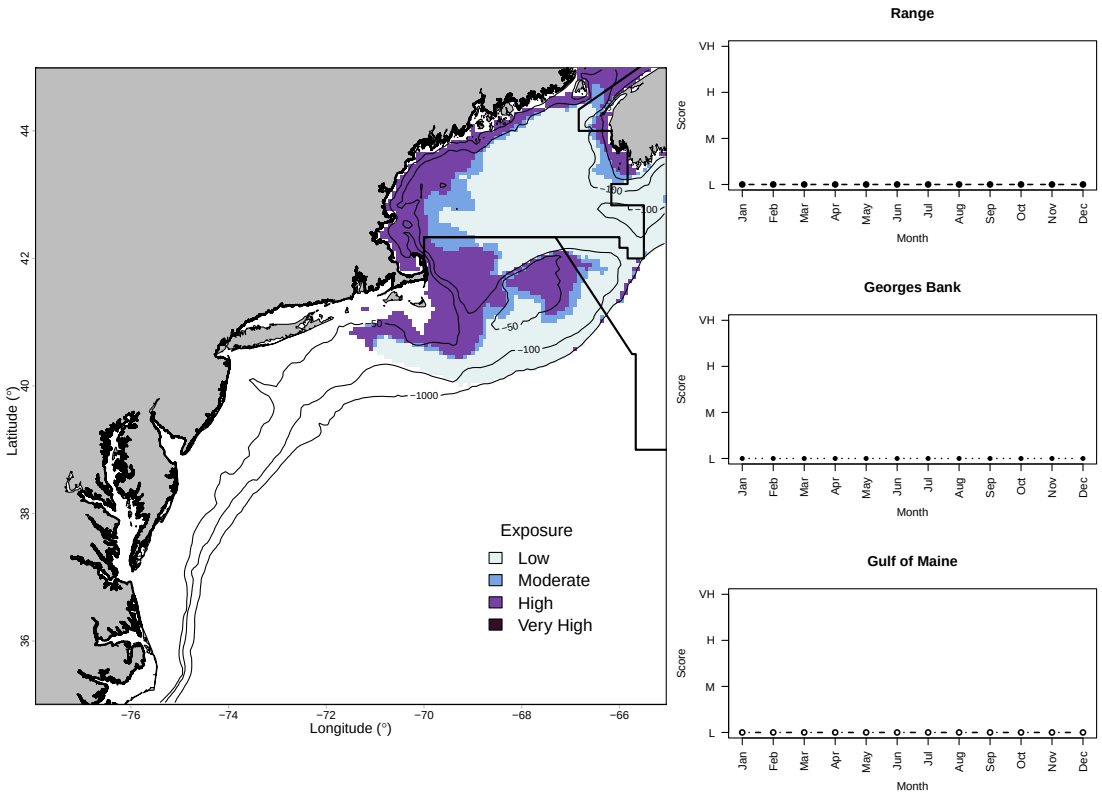


Figure 11: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Haddock distributions

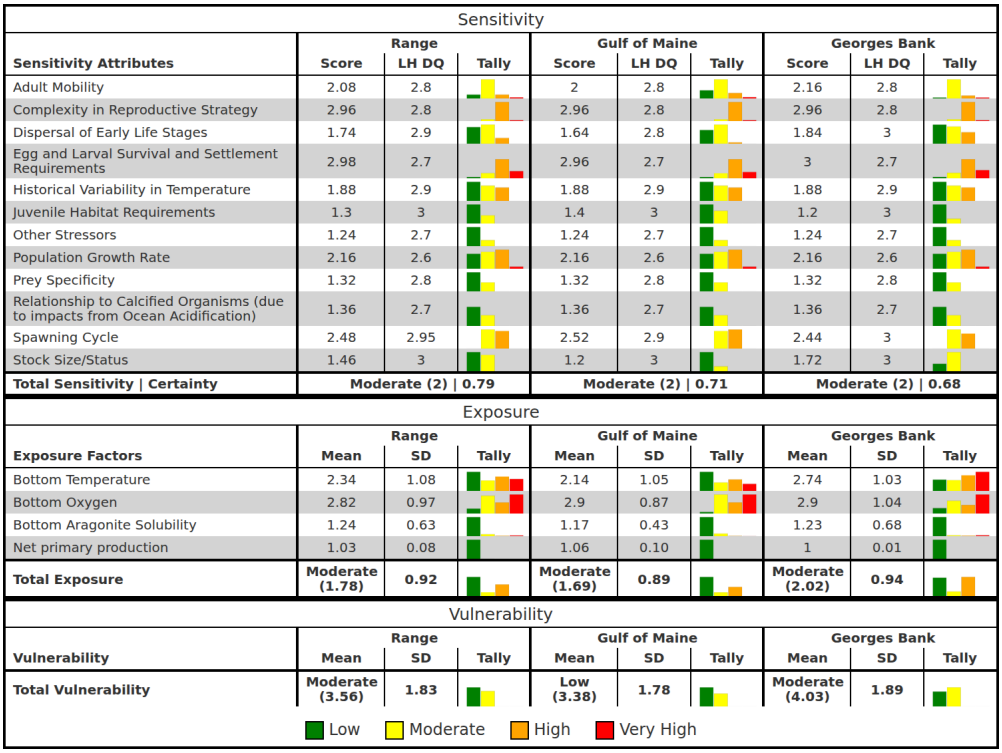


Figure 12: As in Figure 1 for Haddock, but with only significant Exposure factors and the resulting Vulnerability Scores.

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